

Project Design Phase-II Technology Stack (Architecture & Stack)

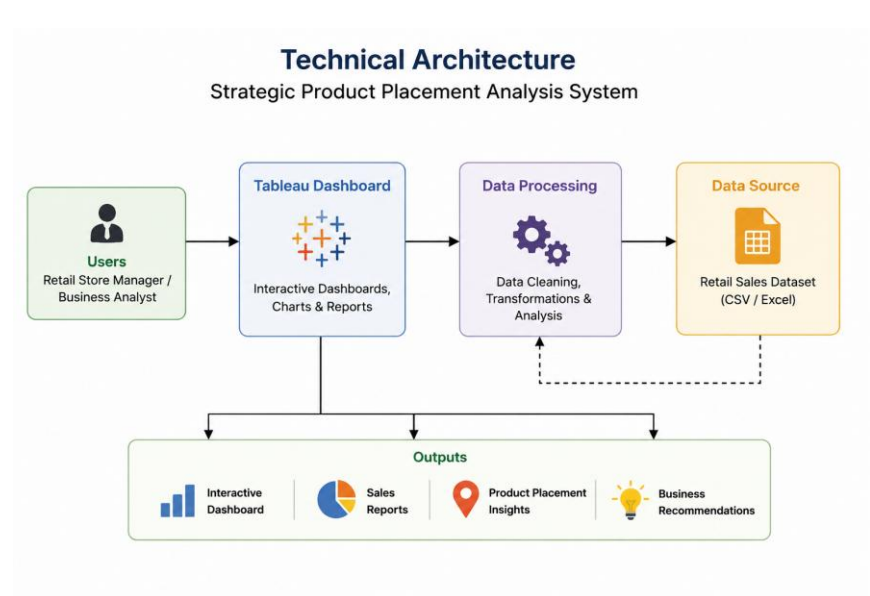
Date	29 June 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- ☐ Import and validate the retail sales dataset before analysis.
- ☐ Clean and preprocess the data to ensure accuracy and consistency.
- ☐ Build interactive Tableau dashboards with filters and visualizations.
- ☐ Analyze product positioning, pricing, promotions, and customer behavior.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard for business users	Tableau Desktop
2.	Data Import	Import retail sales dataset	CSV / Excel
3.	Data Preparation	Clean and organize the dataset	Tableau Prep / Tableau Data Source
4.	Data Analysis	Analyze sales, pricing, promotions, and customer behavior	Tableau Calculated Fields
5.	Visualization	Interactive charts and dashboards	Tableau Desktop
6.	Data Storage	Store retail sales dataset	Local CSV File
7.	File Storage	Store project files and dashboard	Local File System
8.	External API-1	Not Applicable	Not Applicable
9.	External API-2	Not Applicable	Not Applicable
10.	Machine Learning Model	Not Used	Not Applicable
11.	Infrastructure (Server / Cloud)	Local Computer	Windows PC with Tableau Desktop

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Dataset handled using Tableau features	Tableau Desktop
2.	Security Implementations	Dataset stored locally and accessed only by authorized users	Local File Security
3.	Scalable Architecture	Dashboard can analyze larger retail datasets by refreshing the data source	Tableau Data Source

S.No	Characteristics	Description	Technology
4.	Availability	Dashboard can be opened anytime on the local system	Tableau Desktop
5.	Performance	Optimized visualizations, filters, and calculated fields provide fast analysis	Tableau Optimization

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>